

## Independent Course Study

# Predictive Modeling of U.S. Housing Markets: Combining Geographic, Economic, and Environmental Factors

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### Abstract -

This project focuses on predicting housing prices throughout the United States by integrating comprehensive spatial, demographic, economic, and environmental datasets through machine learning techniques. Utilizing advanced geospatial analyses, thorough data preprocessing, and feature engineering, the project identifies critical determinants influencing housing prices. Several machine learning algorithms are rigorously tested, optimized, and validated, ultimately achieving robust predictive accuracy. The project highlights the significance of local factors, captured through ZIP code-level analysis, demonstrating the essential role of spatial data in predictive modeling.

### Introduction -

The United States housing market is characterized by significant fluctuations influenced by diverse geographic, economic, and temporal factors. Housing prices not only reflect the economic health and stability of regions but also directly impact consumer spending and overall economic growth. Economic variables, including median household income, unemployment rates, interest rates, inflation, and market liquidity, significantly influence housing demand and price dynamics. For instance, rising incomes or lower unemployment typically boost housing demand, while higher interest rates can dampen purchasing power and market activity.

Geocoding, the process of converting addresses and ZIP codes into precise geographic coordinates, enables detailed geospatial analysis, crucial for identifying spatial trends and local market dynamics. Spatial analysis methods, such as geographic clustering and heatmap visualizations, help uncover localized pricing patterns and factors influencing property values, such as proximity to amenities, schools, employment centers, and environmental conditions.

Additionally, time series analysis aids in understanding housing price trends and cyclical behaviors over time, offering predictive insights by recognizing patterns associated with economic cycles, seasonal effects, and market sentiments.

The primary objective of this project is to create reliable predictive models for U.S. housing prices by effectively integrating geographic, economic, and temporal data. By employing advanced modeling techniques, the project aims to enhance predictive accuracy, providing actionable insights for real estate stakeholders.

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### Literature Review -

Modern real estate price prediction has evolved from traditional hedonic regression models to sophisticated machine learning techniques. Early work relied on multiple linear regression (hedonic pricing) to link property attributes to price, but such models require strict assumptions (e.g. normal residuals, no multicollinearity). [12] Recent studies demonstrate that machine learning (ML) models can better capture nonlinear relationships and complex interactions in housing data. [9, 12] For example, Čeh et al. (2018) compared a Random Forest regressor to a classic hedonic multiple regression using 7,407 apartment sales and extensive features (property age, structural traits, plus neighborhood and environmental data).[9] The Random Forest consistently achieved higher accuracy (e.g. better  $R^2$  and lower error metrics) than linear regression, confirming that ML outperforms traditional models in predicting housing prices. [9] This finding underscores the benefit of advanced models for nationwide house price prediction, as in the user's project, where nonlinear effects and interactions (e.g. between home features and location) are present.

Traditional hedonic models have been extended with spatial techniques (e.g. Geographically Weighted Regression) to handle spatial heterogeneity. [9] In modern ML, adding geospatial features and structures can similarly enhance predictions. Soltani et al. (2022) demonstrated this by introducing a spatiotemporal lag feature into ML models for housing prices. [10] Using 428,000 sales across metropolitan Adelaide, they found that tree-based models (Decision Trees, Random Forests) already outperformed linear models, and further that adding a spatial-temporal lag variable (reflecting nearby recent sales) significantly improved prediction accuracy. [10] This ST-lag feature acted as a spatially aware predictor, helping capture neighborhood price trends and interactions over time. [10] The user's project, which performs ZIP code-level geospatial analysis, is aligned with these findings – by incorporating location-based features (ZIP code, coordinates, or neighborhood indices), the model can account for local market conditions and spatial dependencies between nearby properties.

Recent research has integrated demographic and economic indicators into price models to capture broader market drivers. For instance, Mora-García et al. (2022) combined property attributes with neighborhood socio-economic and environmental data in a machine learning model for house prices in Alicante, Spain. [11] Their feature set included geographic coordinates, distances to amenities (schools, parks, coastline), and even an environmental quality metric (NDVI – a vegetation index) for each property. [11] Using data from 2019–2021 (pre- and post-COVID), they compared multiple algorithms – including linear regression and ensemble methods (Random Forest, Extra-Trees, Gradient Boosting) – to identify the best predictor of prices. [11] The ensemble models achieved higher accuracy and exhibited less overfitting than linear regression. [11] This result reinforces two points: (1) incorporating diverse features such as spatial amenities and environmental indices improves model performance, and (2) ensemble ML techniques are effective for handling the variety of inputs. The user's plan to integrate demographic and economic datasets (e.g. income levels, employment rates, interest rates) at the ZIP code level is well-supported – such features often proxy demand and purchasing power, thereby enhancing the model's ability to generalize across different communities. [11]

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## Data -

The project utilized diverse datasets from reputable sources:

- Housing Price Data: Zillow House Price Data, serving as the primary dataset containing historical housing price information at the ZIP code level. [1]
- Unemployment Rates: OECD Unemployment Rate, providing insights into regional economic stability and employment conditions influencing housing demand. [2]
- Gasoline Prices: Energy Information Administration (EIA), representing transportation and fuel costs impacting consumer budgets and housing location preferences. [3]
- Inflation Data: Consumer Price Index (CPI), essential for understanding purchasing power and general economic conditions influencing housing affordability. [4]
- Mortgage Data: National Mortgage Database by FHFA, crucial for assessing mortgage market conditions, lending rates, and home affordability. [5]
- Interest Rates: Federal Funds Rate, directly affecting mortgage rates and home-buying capacity. [6]

## Data Preprocessing

To support a robust and multi-dimensional analysis of U.S. housing markets, we designed a comprehensive preprocessing pipeline that prepared the raw data for subsequent modeling and analysis. This pipeline encompassed several stages, including temporal restructuring, geographic enrichment, integration of macroeconomic indicators, and final data cleaning.

The first stage involved temporal restructuring of the housing dataset. The original format, in which each column represented a date, was transposed so that each row corresponded to a unique monthly timestamp. In this new format, the columns represented individual RegionIDs. This transformation was essential to facilitate time series modeling and analysis across multiple geographic units. In addition, date columns were converted into standardized datetime format to support consistent filtering, aggregation, and time-based computation. Columns containing non-essential information such as RegionName, StateName, and SizeRank were removed to simplify the feature space and reduce noise. To address redundancy and ambiguity in geographic identifiers, we analyzed the presence of duplicate city names. Using a pivot table grouped by RegionName, we identified and logged duplicate entries to ensure that geographic aggregation would remain accurate and unambiguous.

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For geographic enrichment, we utilized the uszipcode library's SearchEngine to map each RegionName and StateName pair to its corresponding ZIP code. This added geographic metadata to the dataset and allowed for enhanced spatial granularity in downstream analysis. Entries that could not be resolved through the ZIP code lookup were stored separately in a dedicated dictionary for unresolved cities. To further support geospatial modeling and visualization, we appended latitude and longitude coordinates to each ZIP code using the GeoPy library and the Nominatim API. These geographic coordinates enabled the integration of the dataset with external geospatial layers and supported interactive mapping. Locations that failed to resolve during this process were tracked and excluded from geospatial visualizations.

The enriched dataset was further augmented by integrating several national-level macroeconomic indicators, each selected for its relevance to housing market dynamics. Inflation data, measured by the Consumer Price Index (CPIAUCNS), was included to reflect changes in purchasing power. Federal Reserve interest rates (FEDFUNDS) were incorporated to capture fluctuations in borrowing costs that directly affect mortgage decisions. Average mortgage rates were merged to provide a direct measure of housing affordability. Unemployment rates were added to reflect the strength of local labor markets and their impact on housing demand. Finally, gasoline prices were integrated as a proxy for economic stress and transportation costs, particularly in suburban and rural housing markets.

Each macroeconomic indicator was aligned temporally with the housing dataset using inner joins on the DATE column. This ensured that all indicators corresponded precisely to the same monthly timestamps and preserved the integrity of time-based modeling. The indicators were then reordered in the dataset for interpretability and ease of access during feature selection and modeling.

The final stage involved cleaning and restructuring the dataset to ensure high data integrity. Rows with missing or unmatched values were removed to prevent distortions during training and evaluation. Additionally, we filtered the dataset to retain only those regions with complete geographic information, including valid ZIP codes and resolved coordinates. The dataset was then merged with ZIP code metadata and reordered such that ZIP codes could serve as the primary geographic identifiers in both modeling and visualization workflows. This cleaned and enriched dataset served as the foundation for the subsequent stages of geospatial and machine learning analysis.

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## Process -

### Geospatial and Temporal Visualization of Housing Market Trends

To explore localized housing price trends across the United States, we developed an interactive dashboard leveraging Holoviews and Panel. This tool enabled dynamic visualization of temporal price fluctuations at the ZIP code level, enhancing the granularity of regional housing market analysis. We began by creating a mapping between RegionIDs (ZIP codes) and their corresponding City and State values using a cleaned reference dataset. Invalid or unmatched ZIP codes were filtered out to maintain consistency. This mapping facilitated more intuitive user interaction by allowing city-level selection in place of raw ZIP codes.

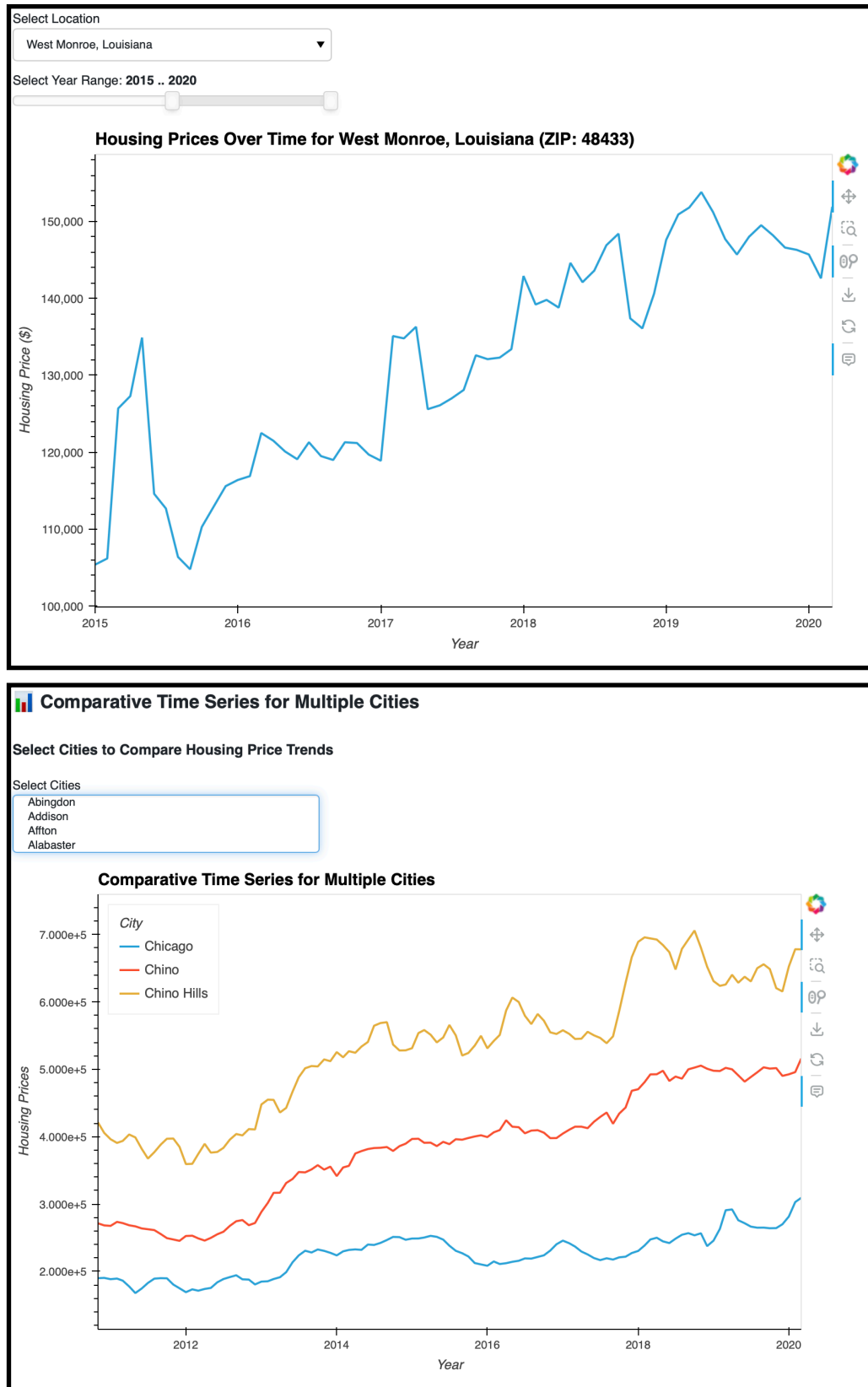
The dashboard featured a dropdown menu for selecting geographic regions and a range slider for filtering data by year. Upon user input, the dashboard dynamically rendered a time series plot using Holoviews' `hv.Curve`, with time on the x-axis and housing prices on the y-axis. Hover tools provided date-specific price values, and formatting enhanced numerical readability. This visualization supported the identification of trends, seasonality, and potential inflection points in housing markets.

To enable comparative regional analysis, we restructured the dataset to represent average housing prices at the city level. The dataset was melted from wide to long format, mapping ZIP codes to cities and aggregating prices by city and date. A multi-selection widget allowed users to view and compare temporal price trajectories across multiple cities simultaneously. This comparative visualization proved valuable in evaluating relative growth patterns, market divergence, and the effects of economic or policy changes across different metropolitan areas.

We further expanded the analysis by constructing an interactive geospatial heatmap using GeoViews and Datashader. The housing dataset was reshaped to associate each ZIP code and year with its corresponding average price and geolocation. Latitude and longitude data were merged from ZIP code metadata, and incomplete records were excluded. The heatmap was rendered using a `DynamicMap`, enabling users to explore spatial price distribution by selecting a year. Each ZIP code was represented as a point on the map, with color intensity reflecting average housing prices. An `Inferno` colormap and OpenStreetMap base tiles provided visual clarity and spatial context. This heatmap visualization facilitated the detection of geographic pricing disparities and highlighted key regions exhibiting elevated or suppressed housing market activity.

Together, these visual tools supported both temporal and spatial analysis of U.S. housing markets, allowing researchers and stakeholders to examine local dynamics, detect regional outliers, and assess broader market behavior with high spatial precision.

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**Figure 1 : Geospatial and Temporal Visualization of Housing Market Trends**

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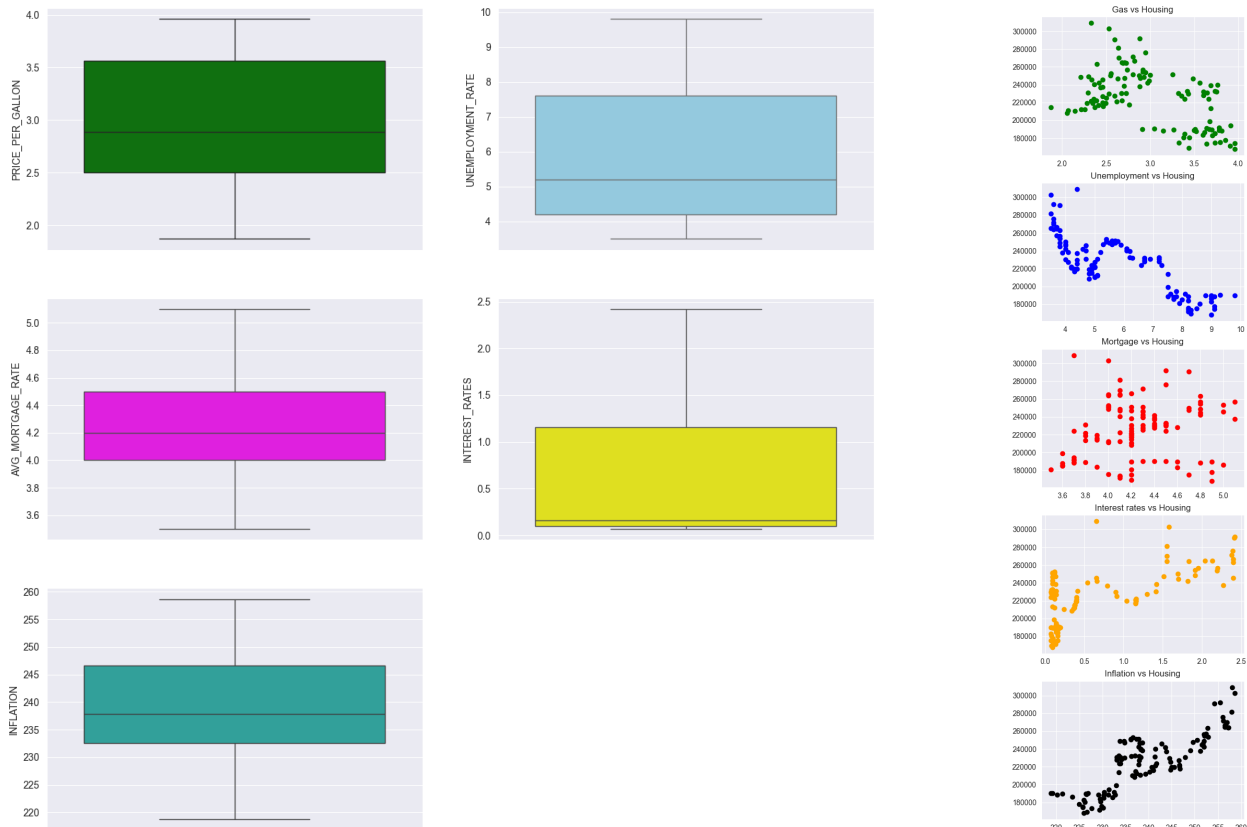
### Modeling Approaches for Housing Price Prediction

To develop a predictive framework for estimating housing prices at the ZIP code level, we evaluated three different supervised learning algorithms: Multiple Linear Regression (MLR), Random Forest Regression (RFR), and a Deep Learning Neural Network. Each model was selected based on its unique strengths and suitability for handling structured economic and geographic data, and evaluated based on predictive accuracy and interpretability.

### Multiple Linear Regression (MLR)

We first employed Multiple Linear Regression as a baseline model due to its simplicity and transparency in estimating linear relationships. MLR assumes that the dependent variable, in this case housing price, can be modeled as a linear combination of independent predictors. The features selected included economic variables such as gas prices, unemployment rate, average mortgage rate, interest rates, and inflation (CPI).

Initial exploratory data analysis using boxplots showed no significant outliers in the dataset, allowing us to proceed without outlier removal. Scatter plots were then used to assess linearity between the predictors and the target variable. This analysis revealed that while most features displayed a linear trend with housing prices, mortgage rates did not exhibit a meaningful linear relationship and were subsequently excluded from the MLR model.



**Figure 2 : BoxPlot Representation for detecting outliers and plots for checking linearity**

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After splitting the dataset into training and test sets using a 75:25 ratio, the model was trained using Scikit-learn's `LinearRegression` class. The resulting model captured approximately 71% of the variance in housing prices ( $R^2 = 0.71$ ), with an adjusted  $R^2$  of 0.66. Performance metrics such as RMSE (16.74) and MAE (14.53) indicated moderate prediction error. To enhance interpretability, we further validated the model using the OLS regression from Statsmodels, which confirmed that inflation and interest rates were statistically significant predictors. The regression diagnostics also showed minimal autocorrelation and manageable multicollinearity, making the MLR model a reliable benchmark.

OLS Regression Results

|                   |                  |                     |          |
|-------------------|------------------|---------------------|----------|
| Dep. Variable:    | y                | R-squared:          | 0.739    |
| Model:            | OLS              | Adj. R-squared:     | 0.726    |
| Method:           | Least Squares    | F-statistic:        | 56.01    |
| Date:             | Thu, 06 Mar 2025 | Prob (F-statistic): | 2.61e-22 |
| Time:             | 13:42:19         | Log-Likelihood:     | -353.23  |
| No. Observations: | 84               | AIC:                | 716.5    |
| Df Residuals:     | 79               | BIC:                | 728.6    |
| Df Model:         | 4                |                     |          |
| Covariance Type:  | nonrobust        |                     |          |

|                   | coef      | std err | t      | P> t  | [0.025   | 0.975]  |
|-------------------|-----------|---------|--------|-------|----------|---------|
| const             | -510.1185 | 219.639 | -2.323 | 0.023 | -947.299 | -72.938 |
| PRICE_PER_GALLON  | -2.8872   | 6.672   | -0.433 | 0.666 | -16.168  | 10.393  |
| UNEMPLOYMENT_RATE | -1.3408   | 4.842   | -0.277 | 0.783 | -10.979  | 8.297   |
| INTEREST_RATES    | -11.9757  | 4.957   | -2.416 | 0.018 | -21.843  | -2.109  |
| INFLATION         | 3.1747    | 0.870   | 3.649  | 0.000 | 1.443    | 4.906   |

|                |        |                   |          |
|----------------|--------|-------------------|----------|
| Omnibus:       | 10.069 | Durbin-Watson:    | 1.969    |
| Prob(Omnibus): | 0.007  | Jarque-Bera (JB): | 5.562    |
| Skew:          | 0.443  | Prob(JB):         | 0.0620   |
| Kurtosis:      | 2.103  | Cond. No.         | 2.89e+04 |

...

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.89e+04. This might indicate that there are strong multicollinearity or other numerical problems.

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**Figure 3 : OLS Regression Results for Multiple linear regression**



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### Random Forest Regression

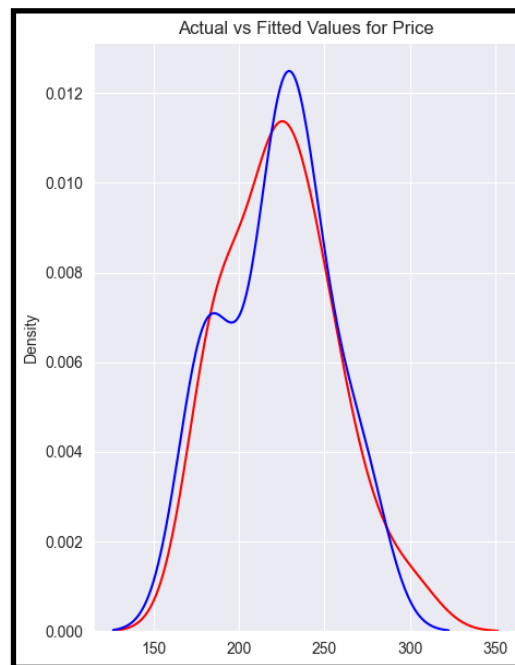
We then implemented Random Forest Regression as a non-linear ensemble method capable of capturing complex feature interactions that MLR may overlook. Random Forest builds multiple decision trees during training and averages their predictions, reducing variance and improving robustness. The model achieved a significant improvement in predictive accuracy compared to MLR. It produced a lower RMSE of 9.13 and a much smaller MAE of 6.92, with an estimated accuracy of 96.89%. Visualization of actual vs. predicted values confirmed that the model was able to closely approximate true housing prices across the test set.

However, a notable limitation of Random Forest is its inability to extrapolate beyond the range of training data. For instance, when asked to predict housing prices for economic conditions not present in the training set, the model produced values strictly bounded within observed outcomes, failing to account for future economic shifts. This constraint, while acceptable for current-state analysis, may limit the model's utility for forward-looking forecasts under extreme or novel scenarios.

|   | Actual | Predicted |
|---|--------|-----------|
| 0 | 216.8  | 224.5406  |
| 1 | 186.3  | 186.1294  |
| 2 | 231.9  | 230.7926  |
| 3 | 219.0  | 221.2337  |
| 4 | 190.0  | 174.7943  |

Mean Absolute Error (MAE): 6.92  
Mean Squared Error (MSE): 83.28  
Root Mean Squared Error (RMSE): 9.13

Mean Absolute Error: 6.92 degrees.  
Accuracy: 96.89 %.



**Figure 4 : Results (RF) and difference between actual and predicted value**

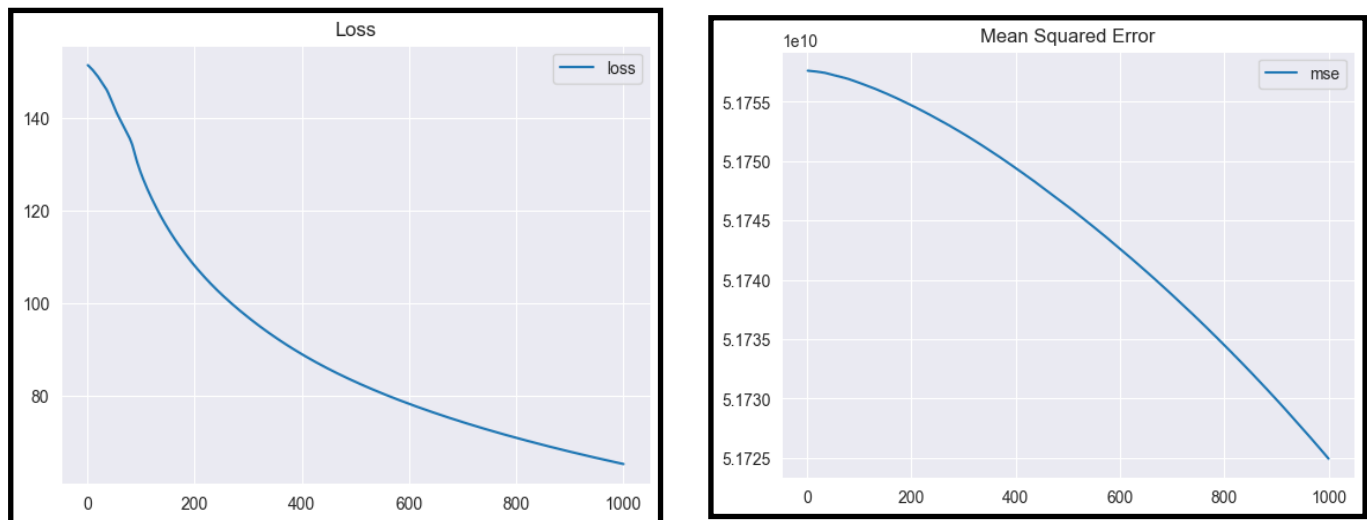
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### Deep Learning Model (Neural Network Regression)

To assess whether a more flexible, non-linear approach could outperform traditional models, we implemented a deep learning regression model using a feedforward neural network built in TensorFlow Keras. The network architecture consisted of an input layer corresponding to five normalized economic features, a hidden layer with ten neurons and ReLU activation, and a single output neuron for predicting continuous housing price values.

Feature scaling using `StandardScaler` was applied prior to model training to ensure that all input variables were within the same numerical range, a critical requirement for gradient-based optimization. The model was compiled with the Adam optimizer and trained for 1000 epochs using mean squared logarithmic error as the loss function, which penalizes large underestimations more than overestimations.

While training loss decreased steadily, the model's test performance was unsatisfactory. It produced an extremely high mean squared error of over 51 billion, suggesting overfitting or insufficient generalization. This underperformance can be attributed to a combination of factors, including a limited dataset, insufficient feature diversity, and a lack of hyperparameter optimization. Despite the theoretical capacity of neural networks to model highly non-linear functions, in this context the deep learning model proved less effective than simpler models due to the scale and structure of the input data.



```
1/1 - 0s - 129ms/step - loss: 64.8790 - mse: 51391447040.0000  
Loss: 64.87895202636719, MSE: 51391447040.0
```

*Figure 5 : Loss during training and Mean Squared Error*

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## Results & Insights:

This study explored the predictive modeling of U.S. housing prices using multiple machine learning techniques, each selected for its unique strengths. Beginning with Multiple Linear Regression (MLR), we ensured key assumptions such as linearity and absence of outliers were met, resulting in a well-performing and interpretable model with an  $R^2$  of 0.71 and low error margins. The use of Ordinary Least Squares (OLS) provided deep statistical insights and validation of model robustness.

Random Forest Regression proved highly accurate (~96%) and excelled in modeling complex interactions within the dataset. However, it fell short in extrapolating to unseen or out-of-range data, as demonstrated in the ZIP code 10001 case, making it less suitable for predictions involving future price increases.

Deep Learning offered a powerful framework but was limited by the dataset size and architectural choices. Despite improving loss over epochs, the model exhibited high variance and less stability. Moreover, the opacity of its internal workings posed challenges for interpretability and deployment.

- **MLR Model Insight:**

Despite the Random Forest's high accuracy, the MLR model provided superior interpretability and better extrapolation capabilities, especially evident when predicting values beyond the training data's range.

- **Random Forest Limitation:**

RF predictions were constrained within the range of training target values, limiting generalization. For ZIP code 10001, RF predicted \$471,273, significantly below MLR's \$596,503 estimate and below the training set's max value (\$575,100).

- **Deep Learning Model:**

Demonstrated the potential to learn complex patterns. However, due to the relatively small dataset, results indicated diminishing returns beyond 1000 epochs. Future improvements may include regularization and dropout to further reduce loss.

## Conclusions:

Multiple Linear Regression model was identified as the most appropriate and balanced solution offering a combination of predictive accuracy, transparency, and flexibility. Its capability to extrapolate and integrate with user-friendly tools such as a GUI enhances its practical value. This conclusion underscores that the best-performing model is not necessarily the most complex, but the one that aligns best with data characteristics, project goals, and interpretability needs.

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